Reading Both Halves of *Imvo*

Team project outline — COMS4054A/7062A/7066A NLP Project 2: Language in Culture and Responsible NLP

Working draft, 13 October 2026 · Submission due **27 October 2026, 17:00**

One-line summary. We place the isiXhosa and English sections of each *Imvo Zabantsundu* issue into one shared embedding space and test whether their content differs by audience. We then modernise the spelling of the isiXhosa, repeat the identical experiment, and measure how that preprocessing choice changes the conclusion: **what are the consequences of modernising historical text?**

1. Research question and hypotheses

RQ. When *Imvo*'s isiXhosa and English sections from the same issue are placed in a shared multilingual embedding space, how distinguishable is their *content*, and how does modernising the 1880s isiXhosa spelling change that answer?

- **H1 (two audiences).** After the language signal is removed, a classifier predicts the section of a paragraph above the *translated-pair floor* (Section 4, Trap 1), i.e. the sections carry different content.
- **H2 (modernisation is not neutral).** Modernising the isiXhosa spelling changes the measured content gap (in either direction) by more than run-to-run variation.
- Null results are acceptable and reportable; the marks are for method and interpretation, not for confirming H1/H2.

2. Why this fits the brief

Brief requirement	How we meet it
Use the isiXhosa Intellectual Traditions data	Imvo Zabantsundu, 1884–1894, language-split text files
Language in culture	One editor, two audiences (isiXhosa readers vs colonial English readers), same week
AI fairness / responsible NLP	English-first analysis erases isiXhosa-only content; 'modernising' text is a hidden preprocessing choice that can change conclusions and overwrites authors' own forms
Train/validation/test + tuning	Splits by issue; classifier, pairing threshold and normaliser threshold all tuned on validation
Report limitations and failed runs	Trap analysis, OCR noise, normaliser error rates are reported explicitly

3. Data and pilot evidence

- **Corpus:** `TxtLangFiles1870-1894_TB` (XhTxt / EnTxt per page) plus `LanguageSplitTracker70-94.csv`. 511 of 517 Imvo issues contain at least 1,000 words in each language (about 4.0M isiXhosa and 1.5M English words in total).
- **Spelling evidence:** the archive writes *Tixo* (about 3,900 occurrences) and never modern *Thixo*; *ute, tetayo, etu, apa, Rulumeni* appear where modern spelling has *uthe, thethayo, ethu, apha, uRhulumente*; prefixes are written apart (*e Palamente*).
- **Pilot (issue 1887-06-01, 105 isiXhosa and 28 English chunks of ~60 words):** LaBSE best-match similarity median 0.40 vs 0.25 for random pairs; top matches (0.54–0.67) were genuine translations (an obituary, the Cala magistrate's report on Timothy Mmango, the franchise petition to Mr Frost MLA, the subscription notice). LaBSE is usable on this text.

4. Method

Stage 0 — Preparation

- Clean OCR artefacts: join line-break hyphens, normalise whitespace, drop mastheads and repeated boilerplate.
- Chunk each page into ~60-word windows; keep issue date, page and language as metadata.

- Split **by issue** (70/15/15, stratified by year) so no issue appears in more than one split.

Stage 1 — Two audiences, raw text

- Embed all chunks with LaBSE (sentence-transformers).
- **Find translated pairs** within each issue: mutual nearest neighbours above a similarity threshold. Tune the threshold on a hand-labelled set of candidate pairs (about 150 validation / 150 test); report precision and recall.
- **Section classifier:** logistic regression on embeddings (regularisation C tuned on validation); report macro-F1 with bootstrap confidence intervals over issues.

Stage 2 — Modernise the isiXhosa

- **Rule-based normaliser:** generate candidates for each historical word (restore aspiration *t→th, p→ph, k→kh, r→rh*; join separated prefixes) and accept a candidate only if it is attested in a modern isiXhosa word list and the original is not. Minimum-frequency threshold tuned on validation.
- **Normaliser check (required):** about 200 hand-checked words split into validation and test; report precision and recall so that effects can be attributed to modernisation rather than normaliser errors.
- Re-embed the normalised isiXhosa; English is unchanged.

Stage 3 — Compare raw vs modernised

- Repeat Stage 1 exactly (same splits, same seeds, re-tuned hyperparameters).
- Report the change in: classifier macro-F1 above the floor, translated-pair floor itself, number of pairs found, mean pair similarity, LaBSE tokens per word, and the share of words missing from the modern word list.

Trap fixes built into the design

Trap	Problem	Fix
1. Trivial classifier	In Imvo the section *is* the language. Embeddings still encode language identity, so a raw classifier scores ~100% by detecting language, saying nothing about content.	(a) Remove the language signal by subtracting each language's mean embedding (mean-centring). (b) Use translated pairs as a **control**: they share content, so classifier accuracy on them is the language/spelling floor. Only accuracy **above** the floor counts as a content difference. (c) Report the uncentred classifier as a sanity check only.
2. Modernisation changes two things	A drop in accuracy after modernising could mean the content now looks more alike, *or* only that the isiXhosa looks less like old isiXhosa (weaker language signal), *or* normaliser mistakes.	Compare the change in the floor with the change in the above-floor gap: if both move equally it is a spelling/language effect; if the gap moves it is a change in perceived content. Normaliser precision/recall rules out error-driven effects.

5. Metrics

Metric	Answers
Macro-F1 (classifier), above the translated-pair floor	H1: do the sections differ in content?
Δ floor and Δ gap, raw vs modernised, with bootstrap CIs over issues	H2: what did modernising change, and why?
Pair-finding precision/recall on labelled pairs	Is the translated-pair control trustworthy?
Normaliser precision/recall; out-of-vocabulary rate; tokens per word	Did modernising work, and what did it do to the tools?

6. Roles

Placeholder names; reassign if the group is smaller. Everyone writes, but each person owns one section of the abstract.

Role	Owns	Abstract section
[Member A] Data & normaliser lead	Stage 0 pipeline, splits, modern word list, Stage 2 normaliser	Data, Method (Stage 2)

Role	Owns	Abstract section
[Member B] Modelling lead	LaBSE embeddings, pair finding, classifier + tuning, mean-centring, Stage 3 runs	Method (Stages 1, 3), Results
[Member C] Evaluation & ethics lead	Hand-labelled pairs and normaliser check, bootstrap CIs, figures, ethics checklist + AI statement	Introduction, Background, Discussion

Hand-labelling is easier if a team member reads isiXhosa; if none does, say so as a limitation and rely on names, numbers and dates shared between sections to judge pairs.

7. Timeline

Dates	Milestone	Owner
Mon 13 – Tue 14 Oct	Contribution statement submitted (due ~now); everyone reads Schoots et al. (2023); repo set up	All
Wed 15 – Thu 16 Oct	Stage 0 pipeline and splits; modern word list sourced; pair labelling starts	A, C
Thu 16 – Sat 18 Oct	Stage 1 on raw text, including mean-centring and the floor	B
Fri 17 – Sun 19 Oct	Stage 2 normaliser + hand check	A, C
Mon 20 – Tue 21 Oct	Stage 3 comparison, bootstrap CIs, final figures	B, C
Wed 22 – Sat 25 Oct	Write the 2-page abstract (most time on Discussion, 30% of marks); internal review	All
Sun 26 Oct	NeurIPS ethics checklist, Faculty AI ethics statement, README.txt, requirements.txt, clean rerun from scratch	C, A
Mon 27 Oct	Buffer; submit before 17:00	All

8. Risks and mitigations

Risk	Mitigation
Few true translated pairs, so the floor is noisy	Pool across all 511 issues; report CIs; fall back to entity/number overlap as a weak pairing signal
Normaliser over-corrects (e.g. adds an *h* where none belongs)	Accept only lexicon-attested candidates; tune threshold; report errors
Modern word list biased to one register (e.g. government or Wikipedia text)	State the source; if time allows, compare two sources
OCR noise dominates differences	Report OOV rate per year; check results hold on cleaner years
Over-reading intent ('Jabavu hid X')	Claim only measured differences; interpretations framed as hypotheses

9. Must verify before writing

Fabricated or wrong facts are heavily penalised. Nothing below goes into the abstract until checked against a source.

- A citable source on how isiXhosa orthography changed (aspiration marking, word division) and when.
- The mean-centring reference (Libovický et al., 2020, on language neutrality of multilingual representations): confirm details.
- Licence and provenance of the modern isiXhosa word list and any parallel data.
- Every historical claim about Imvo, Jabavu and the 1880s franchise, checked against Schoots et al. (2023) or another scholarly source.

10. Submission checklist

- 2-page extended abstract (PDF, Moodle template); references and contribution statement on extra pages.
- NeurIPS 2024 ethics checklist appended after references (project is not marked without it).
- Faculty of Science AI Ethics Statement.
- Full code, requirements.txt, README.txt with run instructions.